Power envelopes, exact defects, and cycle restrictions for the Juggler map

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Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}. \end{cases}$$

It is conjectured that every positive integer eventually reaches 1. We do not prove that conjecture.

We develop an exact finite-word calculus for the Juggler map. For every parity word w realized at n, the endpoint satisfies the power envelope

$$J^{|w|}(n)^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

Hence $3^{\#O(w)} < 2^{|w|}$ forces $J^{|w|}(n) < n$ whenever $n \geq 2$. Local floor remainders lift through later exponents to an exact global defect. We prove its identity, vanishing classification, and composition law: concatenation is a two-term power-gap, and zero defect gives precisely the rigid monochrome towers. Exact inverse cells then yield necessary cycle restrictions, including superquadratic prefixes from a cycle minimum to later even states, and a small-cycle census: the Juggler map has no nontrivial cycle of length at most six. Length seven remains open.

As a secondary application, every even start and every odd start whose first image is even has a uniform one- or two-step descent certificate. A classical discrepancy estimate

$$\left|S_O(N)\right| \ll N^{5/6}, \qquad S_O(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} (-1)^{\lfloor n^{3/2} \rfloor},$$

implies that this uniform short-certificate class has density 3/4. It is neither a density of all descent certificates nor a density of starts that reach 1.

All finite-word statements are conditional on the realized itinerary. The remaining pointwise question is whether almost every odd-to-odd start has some finite descent; no such theorem is proved here.

1. Introduction

The Juggler sequence was introduced by Pickover [1]; see also OEIS A094683 [2]. Universal arrival at 1 remains open.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \qquad O(n) = \lfloor n^{3/2} \rfloor.$$

A word of length k with o odd letters has ideal exponent $3^o/2^k$. Floors are applied after every letter, and a word is available only when the orbit realizes those parities. The paper records the exact finite-word comparison, the compositional slack, the uniform short certificates, and the density of the class those certificates cover.

The main contribution is the exact power-envelope and defect calculus, together with its inverse-cell and cycle consequences, chief among them a small-cycle census: no nontrivial cycle has length at most six. The short-certificate density is a secondary corollary. We do not prove a Collatz theorem or transfer Collatz stopping-time results to J.

1.1 Verification convention

Theorems in Sections 2–4 are proved both below and in Lean under the names listed at the start of each section. Theorem 5.1 is an ordinary human proof and is not formalized. Proposition 4.4 is an exact Python-integer census, not a theorem about an infinite set. The four-block chain named in Section 6 is Lean-certified (`four_block_pe_1999`); the failed reductions there are exact-integer counterexamples, not estimates. No finite computation is used as a termination proof.

1.2 Related maps

The nearest published comparison is the Collatz problem. Lagarias [3,4] surveys parity words and almost-all statements short of totality. Terras [5] and Everett [6] proved that almost every positive integer has a finite Collatz stopping time; Tao [7] later showed that almost all Collatz orbits attain almost bounded values. Those results are methodological cousins, not theorems about J. The Juggler analogue of Terras would be an almost-all descent theorem on the odd-to-odd class.

Crandall [8] and Matthews–Watts [9] treat piecewise-affine Hasse–Syracuse maps; the Juggler branches are floor powers, not affine. Prasad and Prasad [12] estimate excursion and stopping constants for juggler-like random walks; Section 6 keeps that comparison descriptive.

2. Finite words, envelope, and defect

Let $\mathcal{B} = \{E, O\}$. A finite word $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the orbit of n are exactly the letters of w. Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters.

The identities in this section are formalized in Lean (`follows_iff_word`, `image_eq_iterate`, `image_append`, `image_monotone_of_follows`, `power_bound_word`, `power_bound_contracts`, `global_defect_identity`, `global_defect_eq_zero_iff_localsTight`, `global_defect_append`, `global_defect_eq_zero_implies_monochrome`, `power_bound_eq_iff_extremal`, `image_eq_start_defectRati` `one_plus_eta_lt_succ_sq`). The proofs below are the ordinary integer arguments.

Theorem 2.1 (fixed-word monotonicity). If $n \leq m$ and both realize w, then $J^{|w|}(n) \leq J^{|w|}(m)$.

Proof. Induct on w. The empty word is immediate. For the induction step, the two current states realize the same next letter. On the even branch, monotonicity follows from monotonicity of the integer square root; on the odd branch it follows from monotonicity of $x \mapsto x^3$ followed by the integer square root. Apply the induction hypothesis to the prefix and then the appropriate branch inequality. □

The realizing set of a fixed word need not be an interval.

Theorem 2.2 (finite-word power envelope). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

Proof. Write $k = |w|$ and $o = \#O(w)$. The empty word is the equality $n \leq n$. Suppose the bound holds at a realized prefix ending at x, and the next letter is realized.

If the next letter is even, then $J(x) = \lfloor \sqrt{x} \rfloor$, so $J(x)^2 \leq x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{k+1}} = (J(x)^2)^{2^k} \leq x^{2^k} \leq n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x) = \lfloor x^{3/2} \rfloor$, so $J(x)^2 \leq x^3$. Therefore

$$J(x)^{2^{k+1}} = (J(x)^2)^{2^k} \leq x^{3 \cdot 2^k} = (x^{2^k})^3 \leq \left(n^{3^o}\right)^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. □

Corollary 2.3 (exponent-gap contraction). If $n \geq 2$, w is realized at n, and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \leq n^{3^{\#O(w)}}$. The exponent gap and $n \geq 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \geq 1$, one has $m < n$. □

The corollary includes familiar contracting blocks such as *OOOEE*. It does not prove that every start realizes some contracting word.

The floor slack is exact, and it is not an additive path sum. For a single branch,

$$x^e = J(x)^2 + \rho(x), \qquad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \le \rho(x) < 2J(x) + 1$. Write

$$\operatorname{gap}(a, \rho, e) = (a + \rho)^e - a^e,$$

so $a^e + \operatorname{gap}(a, \rho, e) = (a + \rho)^e$. If $e \ge 1$, then $\operatorname{gap}(a, \rho, e) = 0$ if and only if $\rho = 0$, and $\rho^e \le \operatorname{gap}(a, \rho, e)$.

Along a realized word w, write $x_0 = n$ and $x_{j+1} = J(x_j)$, and let $\rho_j = \rho(x_j)$. Define a running slack by $D_0 = 0$ and

$$D_{j+1} = \begin{cases} D_j + \operatorname{gap}(x_{j+1}^2, \rho_j, 2^j), & \text{next letter even}, \\ \operatorname{gap}(x_{j+1}^2, \rho_j, 2^j) + \operatorname{gap}(x_j^{2^j}, D_j, 3), & \text{next letter odd}. \end{cases}$$

The *global defect* is $\Delta_w(n) = D_{|w|}$. An even letter keeps the old slack and lifts the new remainder through 2^j. An odd letter first cubes the running slack and then lifts the new remainder. In particular Δ is not the sum of the local remainders.

Theorem 2.4 (global defect identity). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \qquad \Delta_w(n) \ge 0.$$

Theorem 2.2 is the inequality $\Delta_w(n) \ge 0$.

Proof. The empty word is $n = n + 0$. Suppose a realized prefix of length k with odd count o ends at x and satisfies $n^{3^o} = x^{2^k} + D$.

If the next letter is even, then $x = J(x)^2 + \rho(x)$, so

$$n^{3^o} = (J(x)^2 + \rho(x))^{2^k} + D = J(x)^{2^{k+1}} + D + \operatorname{gap}(J(x)^2, \rho(x), 2^k).$$

The odd count is unchanged, and the new slack is the even update.

If the next letter is odd, then $x^3 = J(x)^2 + \rho(x)$. Cubing the inductive identity gives

$$n^{3^{o+1}} = (x^{2^k} + D)^3 = x^{3 \cdot 2^k} + \operatorname{gap}(x^{2^k}, D, 3).$$

The leading term is

$$x^{3 \cdot 2^k} = (J(x)^2 + \rho(x))^{2^k} = J(x)^{2^{k+1}} + \operatorname{gap}(J(x)^2, \rho(x), 2^k),$$

which is the odd update. $\square$

Theorem 2.5 (vanishing). If w is realized at n, the following are equivalent.

1. $\Delta_w(n) = 0$.
2. Every local remainder along w vanishes.

3. $J^{|w|}(n)^{2^{|w|}} = n^{3^{\#O(w)}}$.

In that case w is monochrome. More precisely, either $w = E^k$ and $n = a^{2^k}$ for an even a, or $w = O^k$ and $n = a^{2^k}$ for an odd a. A realized mixed word therefore has $\Delta_w(n) > 0$.

Proof. The identity of Theorem 2.4 gives (1)⇔(3). For (1)⇔(2), use the recurrence. If $e \geq 1$, a power-gap vanishes if and only if its addend vanishes. The even update is a sum of two nonnegative terms, and the odd update is a sum of two power-gaps; either vanishes only when the incoming slack and the new remainder both vanish. Thus $D_{|w|} = 0$ forces $D_0 = 0$ and every $\rho_j = 0$. The converse is immediate.

If every remainder vanishes, an even step satisfies $x_j = x_{j+1}^2$, while an odd step satisfies $x_j^3 = x_{j+1}^2$. In either case x_j and x_{j+1} have the same parity, so the itinerary is monochrome. If $w = E^k$, repeated substitution gives $n = x_k^{2^k}$; writing $a = x_k$, its parity is even. If $w = O^k$, unique factorization applied to $x_j^3 = x_{j+1}^2$ shows that every prime valuation of x_j is divisible by 2. Iterating this valuation relation shows that every prime valuation of $n = x_0$ is divisible by 2^k, hence $n = a^{2^k}$ for an odd a. Conversely, these even and odd towers make every local remainder zero. This is the extremal classification `power_bound_eq_iff_extremal`. □

Theorem 2.6 (composition). If u is realized at n and v is realized at $m = J^{|u|}(n)$, then

$$\Delta_{uv}(n) = \mathrm{gap}(m^{2^{|u|}}, \Delta_u(n), 3^{\#O(v)}) + \mathrm{gap}(J^{|v|}(m)^{2^{|v|}}, \Delta_v(m), 2^{|u|}).$$

In particular $\Delta_v(m)^{2^{|u|}} \leq \Delta_{uv}(n)$. Every local remainder satisfies $\rho_j^{2^j} \leq \Delta_w(n)$.

Proof. Write $o_u = \#O(u)$ and $o_v = \#O(v)$. Theorem 2.4 on u and on uv gives

$$n^{3^{o_u+o_v}} = \left(m^{2^{|u|}} + \Delta_u(n)\right)^{3^{o_v}} = J^{|uv|}(n)^{2^{|uv|}} + \Delta_{uv}(n).$$

The first power expands as

$$m^{2^{|u|} \cdot 3^{o_v}} + \mathrm{gap}(m^{2^{|u|}}, \Delta_u(n), 3^{o_v}).$$

Theorem 2.4 on v at m rewrites the leading term as

$$\left(J^{|v|}(m)^{2^{|v|}} + \Delta_v(m)\right)^{2^{|u|}},$$

which expands as the second gap plus $J^{|uv|}(n)^{2^{|uv|}}$. Comparing the two expressions for $n^{3^{o_u+o_v}}$ yields the identity. The suffix inequality is $\rho^e \leq \mathrm{gap}(a, \rho, e)$ on the second summand. For a local remainder at index j, split w after j letters: the suffix defect is at least ρ_j, and raising through 2^j gives the stated bound. □

Composition is polynomial, not additive: later odd letters raise the prefix slack to the third power, and later letters of either parity raise a suffix slack through $2^{|u|}$. The naive recurrence $\Delta \leftarrow \Delta + \rho$ is false.

Corollary 2.7 (cycle surplus). If w is a cycle word at n, that is $J^{|w|}(n) = n$, then

$$\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$$

exactly (`image_eq_start_defectRatio`). A cycle burns its entire formal surplus in floor losses.

Proof. Theorem 2.4 with $m = n$. Nonnegativity of $\Delta_w(n)$ makes the difference a genuine one. □

The identity does not supply a state-independent positive tax, and none exists: the single-step slack is provably squeezed by the scale of the state. For one realized letter at x with branch exponent $e \in \{1, 3\}$,

$$x^e < (J(x) + 1)^2$$

(`one_plus_eta_lt_succ_sq`), so the relative slack $1 + \eta = x^e / J(x)^2$ satisfies

$$\eta < \frac{2}{J(x)} + \frac{1}{J(x)^2},$$

which tends to 0 as the state grows. A uniform per-step tax is therefore impossible. The recorded extreme is an OOE block at $n = 180370579261640036336071806107777 \approx 1.80 \cdot 10^{32}$ whose relative slack satisfies $0 < q < 10^{-30}$; this is an exact-integer computation, not an estimate. The inequality $\Delta_w(n) > n^{3^{\#O(w)}} - n^{2^{|w|}}$ on a formally expanding word is exactly $J^{|w|}(n) < n$, so it does not forbid mixed expanding prefixes.

3. Inverse cells and cycles

The exact one-step fibers are

$$J(n) = q \iff q^2 \le n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \le n^3 < (m+1)^2 \quad (n \text{ odd}).$$

An odd fiber contains at most one integer (`odd_cell_unique`). Indeed, suppose $a < b$ lie in the same odd cell indexed by m. Then $(a+1)^3 \le b^3 < (m+1)^2$ and $m^2 \le a^3$. Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a+1)^3 - a^3 < (m+1)^2 - m^2 = 2m + 1,$$

and hence $3a^2 \le 2m$. If $a = 0$, then $m = 0$ and the displayed strict inequality is already impossible. If $a > 0$, squaring and using $m^2 \le a^3$ gives $9a^4 \le 4m^2 \le 4a^3$, contradicting $4a^3 < 9a^4$. An even fiber is a parity-restricted square interval and may contain many predecessors.

A nonempty realized word w with $J^{|w|}(n) = n$ is a cycle word.

The identities in this section are formalized in Lean (`cycle_word_formally_expanding`, `cycleMin_not_end_odd`, `square_scale_superquadratic`, `cycleMin_to_even_superquadratic`, `no_cycle_word_oooeoe`, `no_cycle_word_ooooee`, `no_cycle_word_length_le_six` with components `no_cycle_word_replicate_odd`, `cycleWord_rotateWord`, `cycleWord_exists_even_terminating`, `no_cycle_word_length_four_ends_even`, `no_cycle_word_length_five_ends_even`, `no_cycle_word_ooe`, `no_cycle_odd_run_append_even`, `no_cycle_word_ooeooe`). The proofs below are the ordinary integer arguments.

Theorem 3.1 (cycle restrictions). Let w be a cycle word at $n \ge 2$.

(i) The word is formally expanding:

$$2^{|w|} < 3^{\#O(w)}$$

(`cycle_word_formally_expanding`). A contracting word cannot close a nontrivial cycle.

(ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter (`cycleMin_not_end_odd`).

(iii) A realized word v is *superquadratic* if

$$3^{\#O(v)} \geq 2^{|v|+1}.$$

Any realized path from a start $n \geq 2$ to a state at least n^2 is superquadratic (`square_scale_superquadratic`). On a cycle minimum the path to any later even state is superquadratic (`cycleMin_to_even_superquadratic`). The prefix *OOE* is expanding ($9 > 8$) but not superquadratic ($9 < 16$), so it cannot carry the minimum to square scale.

Proof. For (i), Theorem 2.2 applied to a cycle endpoint gives $n^{2^{|w|}} \leq n^{3^{\#O(w)}}$. Since $n \geq 2$, comparison of exponents gives $2^{|w|} \leq 3^{\#O(w)}$; equality is impossible because the two sides have different prime divisors and $|w| \geq 1$.

Every state on such a cycle is at least 2: once an orbit reaches 1, it remains there and cannot return to a start $n \geq 2$. An even state $x \geq 2$ satisfies $J(x) < x$, so a cycle minimum cannot be even. An odd state $x \geq 3$ satisfies $J(x) > x$, so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let x be the predecessor of a minimum-oriented return n. If the last letter were odd, the odd return cell would give $n^2 \leq x^3 < (n+1)^2$. Minimality gives $x \geq n$, hence $n^3 < (n+1)^2$, impossible for $n \geq 3$; and the minimum is odd, so $n \geq 3$.

For (iii), let a realized word v send n to $y \geq n^2$. Theorem 2.2 gives

$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \leq y^{2^{|v|}} \leq n^{3^{\#O(v)}}.$$

Since $n \geq 2$, comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state y is even, its successor is at least n. Thus $n \leq J(y) = \lfloor \sqrt{y} \rfloor$, so $n^2 \leq y$, and the preceding argument applies to that prefix. □

These restrictions assemble into a complete census of short cycles. Rotation reduces any cycle word to an even-terminating orientation, the expanding filter of Theorem 3.1(i) and threshold arguments eliminate every even-terminating word of length at most six except *OOOEOE* and *OOOOEE*, and those two survivors require an individual argument, which uses the last-even cell together with a coarse lower envelope. That argument is Lemma 3.2; the census is Theorem 3.3. No exclusion of cycles of length seven or more is claimed.

For $n \geq 1$,

$$n < 4 \lfloor \sqrt{n} \rfloor^2,$$

so an even step satisfies $n \leq 4\, J(n)^2$ and an odd step satisfies $n^3 \leq 4\, J(n)^2$. Composing along a realized word v of length k with odd count o gives

$$n^{3^o} \leq D_v\, J^k(n)^{2^k}.$$

The denominator starts at 1 and updates by $D \mapsto D \cdot 4^{2^j}$ on an even letter and $D \mapsto D^3 \cdot 4^{2^j}$ on an odd letter, at step j. In particular $D_{OOO} = 2^{38}$ and $D_{OOOO} = 2^{130}$. If a cycle word ends in an even letter, the last-even cell is $J^{|w|-1}(n) < (n+1)^2$.

Lemma 3.2 (two length-six exclusions). Neither *OOOEOE* nor *OOOOEE* is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 256$, then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed $257^{64} < 2 \cdot 256^{64}$ and $256(n+1) \le 257\,n$, so $(n+1)^{64} < 2n^{64}$. Then $2^{130}(n+1)^{64} < 2^{131}n^{64}$. Since $n \ge 256 = 2^8$, one has $n^{17} \ge 2^{136}$, and therefore $2^{131}n^{64} < 2^{136}n^{64} \le n^{81}$.

For $n < 256$, neither word returns to its start. Lean checks both itinerary-and-return tables by `native_decide` on `Fin 256`; the same mechanism also verifies the finite numerical inequality $257^{64} < 2 \cdot 256^{64}$ used at the cutoff.

Now suppose $n \ge 256$ realizes $OOOOEE$, and write $z = J^4(n)$ for the image after the prefix $OOOO$. The last two even letters give $J(z) < (n+1)^2$ and $z < (J(z)+1)^2$, hence $z < (n+1)^4$. The lower envelope on $OOOO$ is $n^{81} \le 2^{130}z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose $n \ge 256$ realizes $OOOEOE$. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor\sqrt{z_3}\rfloor$, so $z_3 < (y+1)^2$. The lower envelope on OOO is $n^{27} \le 2^{38}z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \ge 257$. Then $(y+1)^3 < 2A^4$: if $y \le A$, this follows from $(A+1)^3 < 2A^4$; if $y > A$, then $y \ge 4$ and $(y+1)^3 = y^3 + 3y^2 + 3y + 1 < A^4 + 4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \le yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. □

Theorem 3.3 (small-cycle census). No word of length at most six is a cycle word at any $n \ge 2$ (`no_cycle_word_length_le_six`). Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

Proof. Rotating a cycle word by one letter moves the base point one step along the orbit and yields another cycle word (`cycleWord_rotateWord`); every state of the cycle is at least 2, because an orbit that reaches 1 stays at 1 and cannot return to a start $n \ge 2$.

If every letter is odd, the start is odd, hence $n \ge 3$, and the odd branch strictly increases there: $J(x) > x$ for odd $x \ge 3$, since $x^3 \ge (x+1)^2$. The orbit ascends strictly and never returns (`no_cycle_word_replicate_odd`). Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle word vE of the same length based at a cycle state $m \ge 2$ (`cycleWord_exists_even_terminating`). It therefore suffices to exclude even-terminating cycle words of length at most six.

By Theorem 3.1(i) a cycle word is formally expanding. No even-terminating word of length one or two is expanding ($2 < 3^0$ and $4 < 3$ both fail). For length three the only expanding candidate is OOE: for $m \ge 5$ the next-square threshold after OO gives $J^2(m) \ge (m+1)^2$, contradicting the last-even cell $J^2(m) < (m+1)^2$; the only smaller odd start is $m = 3$, where $J^2(3) = 11$ is odd while the final even letter requires an even state. For length four the expanding filter leaves only O^3E, and for length five only O^4E; the threshold inherits along appended odd letters, and the same last-even contradiction applies (`no_cycle_word_length_four_ends_even`, `no_cycle_word_length_five_ends_even`).

For length six the filter requires at least four odd letters among the first five, leaving O^5E, $EOOOOE$, $OEOOOE$, $OOEOOE$, $OOOEOE$, and $OOOOEE$. The odd-run threshold excludes O^5E (`no_cycle_odd_run_append_even`). For $OOEOOE$, a minimum-based orientation meets the same next-square threshold at the internal even letter after the prefix OO, with starts below 5 checked directly (`no_cycle_word_ooeooe`). $EOOOOE$ rotates one step onto $OOOOEE$, and $OEOOOE$ rotates two steps onto $OOOEOE$. The two remaining words are excluded by Lemma 3.2. □

The census stops at length six because length seven admits even-terminating expanding words with two internal even letters that none of the recorded thresholds reach; no exclusion at length seven is claimed.

4. Short descent certificates

A start $n \geq 2$ has a *descent certificate* if there exists a realized finite word w with $J^{|w|}(n) < n$, or with image 1. Lean packages this predicate as `FiniteProgress`, an abbreviation of `DescentCertificate`. The four constructors of that type are proof forms of the same predicate, not four different claims. The predicate is existential over all finite words, not only over words of length one or two.

The theorems in this section are formalized in Lean (`even_finiteProgress`, `odd_even_finiteProgress`, `unresolved_is_odd_odd`, `reachesOne_of_all_finiteProgress`, `reachesOne_of_lt_twelve`, `even_lt_sq_twelve_reachesOne`). The proofs below are the ordinary integer arguments. Proposition 4.4 is the exact census of Section 1.1, not a Lean theorem.

Theorem 4.1 (uniform short certificates). Let $n \geq 2$.

1. If n is even, then the one-letter word E is a descent certificate: $J(n) = \lfloor\sqrt{n}\rfloor < n$.
2. If n is odd and $J(n)$ is even, then the two-letter word OE is a descent certificate.

Proof. For (1), $\lfloor\sqrt{n}\rfloor \leq \sqrt{n} < n$ for $n \geq 2$. For (2), the word OE is realized by hypothesis, and

$$J^2(n) = \left\lfloor\sqrt{\lfloor n^{3/2}\rfloor}\right\rfloor \leq n^{3/4} < n$$

for $n \geq 2$. □

Theorem 4.2 (unresolved starts are odd-to-odd). If $n \geq 2$ has no descent certificate, then n is odd and $J(n)$ is odd.

Proof. Contrapositive of Theorem 4.1: an even start and an odd-to-even start each carry a short certificate. □

The converse is false: many odd-to-odd starts descend after a longer word.

If every start above 1 has some descent certificate, ordinary strong induction yields arrival at 1 (`reachesOne_of_all_finiteProgress`). The hypothesis is not proved. A certificate at n only reduces the problem to a strictly smaller positive integer, which may itself be odd-to-odd.

Lean also certifies a finite landing class:

Theorem 4.3 (small residuals). Every $y \in \{1, \ldots, 11\}$ reaches 1. Consequently every even residual strictly below $144 = 12^2$ reaches 1.

Proof. The orbits are finite and merge quickly: $2 \to 1$, $4 \to 2$, $6 \to 2$, $8 \to 2$, $3 \to 5 \to 11 \to 36 \to 6$, $7 \to 18 \to 4$, $9 \to 27 \to 140 \to 11$, and $10 \to 3$. Each chain ends on a start already settled, so every $y \leq 11$ reaches 1. For the second claim, an even $y < 144$ has $J(y) = \lfloor\sqrt{y}\rfloor \leq 11$, which lands in the verified set. (Lean evaluates the same finite orbits in `reachesOne_of_lt_twelve` and `even_lt_sq_twelve_reachesOne`.) □

This enlarges the set of fatal landings for a hypothetical minimal counterexample.

On the complementary odd-to-odd class, first return below the start is frequent at a short horizon, but not automatic.

Proposition 4.4 (odd-to-odd first-return census). For starts $2 \le n \le N$ and horizon 20, write OO for the odd-to-odd starts in that range. The exact census is:

N	#OO	OO first return ≤ 20	all starts, first return ≤ 20
10^3	252	221/252 (0.877)	968/999 (0.969)
10^4	2504	2220/2504 (0.887)	9715/9999 (0.972)
10^5	24984	22379/24984 (0.896)	97394/99999 (0.974)
10^6	249926	223683/249926 (0.895)	973756/999999 (0.974)

At $N = 10^6$, 26,243 odd-to-odd starts have no return below the start in 20 steps. These rows use exact Python integer arithmetic with no size cutoff through step 20; the unresolved count is zero in every row. They are not Lean-certified and do not constitute an almost-all theorem.

5. Ambient discrepancy and the short-certificate class

For odd n write $s(n) = (-1)^{\lfloor n^{3/2} \rfloor}$ and

$$S_O(N) = \sum_{\substack{n \le N \\ n \text{ odd}}} s(n).$$

Let $M(N)$ be the number of odd integers $n \le N$, and let

$$\mathrm{OO}(N) = \#\{n \le N : \ n \text{ odd}, \ J(n) \text{ odd}\}.$$

Then $s(n) = -1$ if and only if $J(n)$ is odd, and

$$S_O(N) = M(N) - 2\,\mathrm{OO}(N).$$

Also $M(N) = N/2 + O(1)$.

The floor sign has an exact fractional-part form. Write $x = \lfloor x \rfloor + \{x\}$. If $\lfloor x \rfloor$ is even then $\{x/2\} < 1/2$; if $\lfloor x \rfloor$ is odd then $\{x/2\} \ge 1/2$. Thus

$$\lfloor x \rfloor \text{ is odd} \quad \Longleftrightarrow \quad \{x/2\} \ge \tfrac{1}{2}.$$

For odd $n = 2r + 1$ set

$$g(r) = \frac{(2r+1)^{3/2}}{2}.$$

Then $s(n) = -1$ if and only if $\{g(r)\} \ge 1/2$. For R points $x_0, \dots, x_{R-1}$ and an interval $I \subset [0, 1)$, define the unnormalized interval discrepancy

$$\mathcal{D}_R(I) = \#\{0 \le r < R : \{x_r\} \in I\} - R|I|.$$

Taking $R = M(N)$, the odd integers at most N are exactly $2r + 1$ for $0 \le r < R$, and therefore $S_O(N) = -2\mathcal{D}_R([1/2, 1))$ for $x_r = g(r)$.

The following two lemmas are classical. The derivative estimate is the $k = 2$ case of Iwaniec–Kowalski [11, Thm. 8.20, Sec. 8.3, pp. 204–213]. The discrepancy inequality is Kuipers–Niederreiter [10, Ch. 2, Sec. 2, Thm. 2.5, pp. 112–115].

Lemma 5.A (van der Corput, second-derivative form). Let f be twice differentiable on an interval of length M, with $\lambda \le |f''| \le \alpha\lambda$ for some $\alpha \ge 1$. Then

$$\Big|\sum e(f)\Big| \ll_\alpha M\lambda^{1/2} + \lambda^{-1/2},$$

where the sum runs over the integers in the interval and $e(t) = e^{2\pi i t}$.

Lemma 5.B (Erdős–Turán). The discrepancy of R points $x_j \in \mathbb{R}/\mathbb{Z}$ against an interval satisfies

$$D \ll \frac{R}{H} + \sum_{h=1}^{H} \frac{1}{h}\Big|\sum_{j=1}^{R} e(hx_j)\Big|$$

for every cutoff $H \ge 1$.

Theorem 5.1 (ambient odd-input discrepancy).

$$|S_O(N)| \ll N^{5/6}.$$

This argument is not packaged in Lean.

Proof. The second derivative is

$$g''(r) = \tfrac{3}{2}(2r+1)^{-1/2}.$$

It is positive and decreasing. Separate the initial term $r = 0$, then partition $1 \le r < R$ into blocks $[M, \min(2M, R))$, where M runs over powers of 2. On such a block $g''(r) \asymp M^{-1/2}$. For the h-th Fourier mode take $f = hg$, so $\lambda \asymp hM^{-1/2}$. Lemma 5.A gives

$$\Big|\sum_{r \asymp M} e(hg(r))\Big| \ll h^{1/2}M^{3/4} + h^{-1/2}M^{1/4}.$$

Lemma 5.B, with $R \asymp M$, bounds that block's contribution to $S_O(N)$ by

$$\frac{M}{H} + M^{3/4}H^{1/2} + M^{1/4}.$$

Here the middle term is $\sum_{h \le H} h^{-1} \cdot h^{1/2}M^{3/4} \ll M^{3/4}H^{1/2}$, and the last term is $\sum_{h \le H} h^{-1} \cdot h^{-1/2}M^{1/4} \ll M^{1/4}$. For bounded M, the trivial estimate is absorbed into the constant. Otherwise choose $H = \lfloor M^{1/6} \rfloor$, which balances the first two terms at $M^{5/6}$. The initial term contributes $O(1)$, and the geometric sum of the block bounds is $\sum_{M \le R} O(M^{5/6}) = O(R^{5/6}) = O(N^{5/6})$. The floor sign is not replaced by a single exponential; the exponential sums are those of the sequence $\{g(r)\}$. □

Corollary 5.2 (short-certificate class). The odd-to-odd starts have natural density 1/4:

$$\big|\mathrm{OO}(N) - N/4\big| \ll N^{5/6}.$$

Equivalently, the starts that admit the uniform one- or two-step certificates of Theorem 4.1 — every even start together with every odd-to-even start — have natural density 3/4.

This is a density of the uniform short-certificate class of Theorem 4.1, not of all descent certificates and not of starts that reach 1.

The exact census through $N = 10^7$ has $\mathrm{OO}(N)/N = 0.2499896$. Through $N = 10^6$ one has $S_O(N) = 146$ and running maximum 256, at $n = 985351$. A spot computation at 10^7 has running maximum 459. The observed growth is much smaller than $N^{5/6}$: the recorded maxima at 10^6 and 10^7 sit near the scale $N^{1/3}$. That empirical scale is an observation, not a theorem.

Theorem 5.1 concerns consecutive source intervals. It supplies no transfer theorem for orbit samples or sparse image sets, so it cannot by itself control the odd-to-odd dynamics after the first step.

6. The remaining gap

Theorem 4.1 and Corollary 5.2 together say that a uniform one- or two-step argument covers a set of density 3/4. The first odd-to-odd image expands, so ordinary strong induction cannot fire on that class. Proposition 4.4 shows that most odd-to-odd starts in a finite window still return below the start inside twenty steps. That is an observation, not Terras's theorem for J.

A mixed-parity heuristic, ignoring floors, gives mean log-log drift $\frac{1}{2}\log(3/4) < 0$. Finite ensembles sit near this value; hard paths are more odd-rich. This agrees qualitatively with the juggler-like random-walk model of Prasad and Prasad [12]. Fair parity is an assumption, not a dynamical theorem, and typical negative drift is not pointwise contraction.

The gap also resists the obvious finite-state attacks, and the failures are recorded, not anecdotal. Lean certifies a chain of four consecutive expanding persistent residual blocks,

$$1999 \to 5169 \to 50093 \to 193753 \to 887471$$

(`four_block_pe_1999`), so no uniform run bound of four or less exists for expanding blocks. Around such hard paths, five natural finite-state reductions fail on exact-integer counterexamples: a uniform slack tax per expanding block fails at the OOE block near $1.80 \cdot 10^{32}$ with relative slack below 10^{-30} (Section 2); a compact state built from the normalized landing position θ fails because θ fills essentially all of $[0, 1]$ on continuations and predicts the next parity no better than a residue class; recognition modulo 2^m fails because every residue class modulo 64 both continues and exits the iterated odd-landing sets; a 2-adic restriction from persistent-expanding history fails because the endpoint 763 has $v_2(y^3 - J(y)^2) = 1$; and the expanding word grammar is not an independent obstruction because, on $n \geq 2$, persistence already implies expansion, so that attack reduces to the tautology $J^{|w|}(n) > n$. Each failed reduction is an exact computation on named witnesses.

The gap is therefore:

> No theorem forces every exact integer state into a contracting prefix. In particular, it is open whether almost every odd-to-odd start has a finite descent certificate.

That is the Juggler form of the Terras question. A sharper ambient discrepancy exponent does not answer it.

7. Software archive

Lean proofs of the exact-arithmetic theorems live at https://github.com/sneakyweasel/balanced_ternary/. The archive is not required to read the arguments above. Theorem 5.1 is a human proof; Lean does not certify it.

From a clone, the review object is

```
cd formal && lake build Problems.JugglerPaper
```


Acknowledgments

I used large language models extensively while drafting and revising the text, organizing companion notes, and as an interactive assistant for Lean statements, tests, and literature records. The models are not authors. Lean theorems and named computations are the certificates for those claims. The

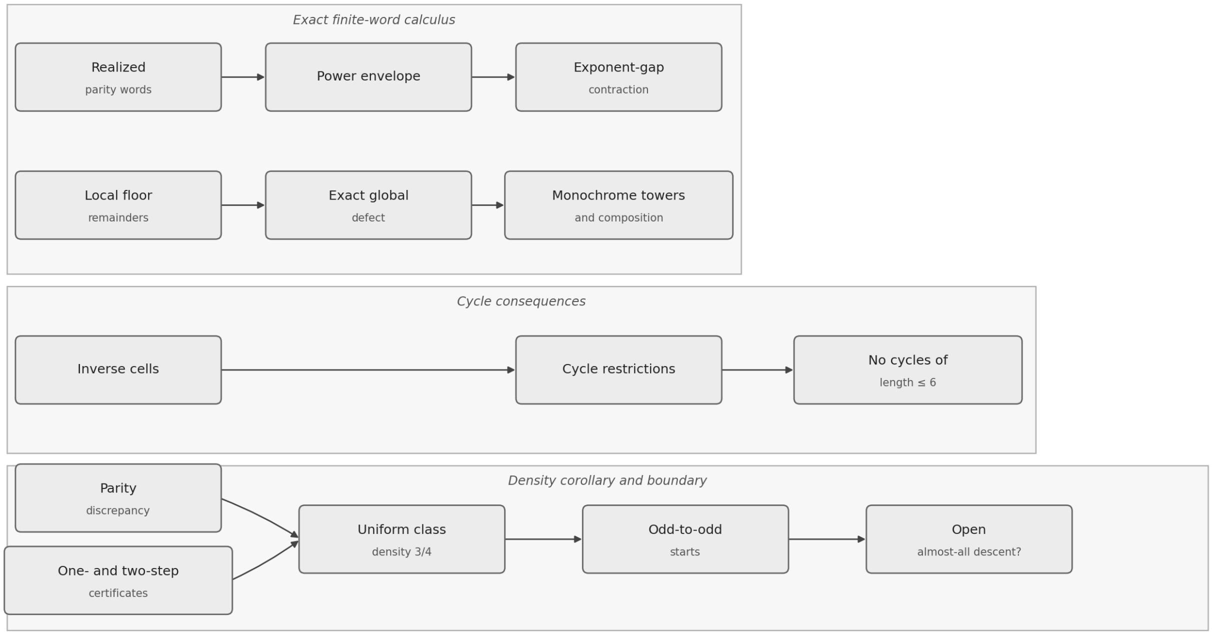

Figure 1: The theorem flow of the note. Exact finite-word identities yield contraction, rigidity, and cycle restrictions; the discrepancy argument counts the uniform short-certificate class but leaves almost-all descent on odd-to-odd starts open.

discrepancy bound (Theorem 5.1) is a human proof using classical analytic inequalities; it is not Lean-certified. I take full responsibility for the contents.

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